

Three-Prime Sign Frustration in a Physical Gram Shell

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Abstract

TPC-291 identified an exact low-residual signed direction for a pair of physical prime components. This paper tests the next compatibility question: can three pairwise cancellation preferences be realized simultaneously? We prove a triangle parity criterion: for three nonzero Gram cross terms, coefficient signs can make all three pair contributions nonpositive if and only if the product of their signs is negative. We also prove the exact three-vector Schur projection formula

$$\min_{\alpha, \beta} \frac{\|g_i - \alpha g_j - \beta g_k\|^2}{\|g_i\|^2} = \frac{\det G_{ijk}}{G_{ii} \det G_{(j,k)}}.$$

An exact-rational replay of the inherited 18-row literal grid evaluates all 5,727 prime triples: 5,718 are sign-frustrated and 9 are anti-alignable, with positive normalized Gram volume in all 5,727 cases. The finite atlas turns pairwise cancellation into a concrete multi-prime obstruction. It does not establish a growing-shell compatibility theorem, an arithmetic L^2 estimate, or a twin-prime result.

1 Position on the twin-prime route

This release remains inside the same literal prime-shell dynamical family. The earlier route separated physical prime components and measured their cross-prime Gram geometry [4, 3, 2]. Diffuse nonnegative weights were shown to face an all-positive coherence wall [1], while TPC-291 proved that an individual coherent pair has an exact signed Schur cancellation direction [5]. Those facts leave a precise question: are the preferred signs for many pair directions mutually compatible?

Three components are the smallest setting in which compatibility can fail by cycle parity. We therefore use the parent physical output without changing the source, interval, kernel, or prime-shell convention. The result is a structural probe on the road map, not a claim that the finite grid represents an asymptotic prime distribution.

2 Physical Gram model

Let $I = I_N$ be the frozen integer interval and let β be the frozen rational source vector. For an odd prime q , height H , and exponent s , write

$$K_{H,s}(h) = \frac{H^{2s}}{(H^2 + h^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The physical deleted-diagonal component is $g_q = A_q \beta$, with

$$(A_q)_{u,t} = q K_{H,s}(u-t) \mathbf{1}_{u \neq t} B_q(u, t).$$

For a shell S , define

$$d_q = \|g_q\|^2, \quad G_{q,r} = \langle g_q, g_r \rangle.$$

All computations below use exact rational arithmetic. Thus a displayed decimal is only a convenient rendering of an exact fraction in the JSON certificate.

3 The compatibility theorem

Theorem 1 (Triangle sign parity). *Let G_{12}, G_{13}, G_{23} be nonzero real numbers. There exist signs $a_1, a_2, a_3 \in \{-1, +1\}$ such that*

$$a_i a_j \operatorname{sign}(G_{ij}) = -1 \quad ((i, j) = (1, 2), (1, 3), (2, 3))$$

if and only if

$$\operatorname{sign}(G_{12}G_{13}G_{23}) = -1.$$

Proof. Multiplying the three required equations makes each a_i occur twice, so the left coefficient factor is 1 and the product of the edge signs must be $(-1)^3 = -1$. Conversely, if the product is -1 , choose $a_1 = 1$, $a_2 = -\operatorname{sign}(G_{12})$, and $a_3 = -\operatorname{sign}(G_{13})$. The first two edges have the desired sign, and the third has

$$a_2 a_3 \operatorname{sign}(G_{23}) = \operatorname{sign}(G_{12}G_{13}G_{23}) = -1.$$

□

Remark 1. *We call the product $+1$ case sign-frustrated. It means that every coefficient-sign assignment leaves at least one positive pair contribution. The theorem is a statement about a signed triangle; it does not optimize coefficient magnitudes or assert that all edges are equally important.*

4 Three-vector Schur residual

Proposition 1 (Three-vector projection identity). *Let g_i, g_j, g_k have Gram matrix G , and suppose the two-vector block $M = G_{(j,k),(j,k)}$ is positive definite. Then*

$$\min_{\alpha, \beta \in \mathbb{R}} \frac{\|g_i - \alpha g_j - \beta g_k\|^2}{G_{ii}} = \frac{\det G}{G_{ii} \det M}.$$

Proof. Put $d_i = G_{ii}$ and $v = (G_{ij}, G_{ik})^T$. Expanding the square gives $d_i - 2v^T c + c^T M c$, where $c = (\alpha, \beta)^T$. The unique minimizer is $c = M^{-1}v$, and the minimum is $d_i - v^T M^{-1}v$. The block determinant identity gives $\det G = \det M (d_i - v^T M^{-1}v)$; division by $d_i \det M$ proves the formula. □

Corollary 1 (Normalized volume). *For three vectors, $\det G / (d_i d_j d_k)$ is the squared volume after normalizing each vector to unit length. It is nonnegative by positive semidefiniteness of a Gram matrix, and is positive exactly when the three vectors are linearly independent.*

The projection formula complements the parity theorem: a small Schur residual says that one vector is close to the span of two others, but it does not say that the signs of all pairwise cross terms can be chosen to lower the same quadratic form.

5 Exact finite protocol

The certificate inherits eight $s = 2$ growth anchors, four $s = 1$ exponent crossover rows, and six $s = 2$ source-control rows, for 18 rows. The prime shells have cardinalities ranging from 3 to 17. Every unordered triple is evaluated, giving

$$\sum_{\text{rows}} \binom{|S|}{3} = 5727.$$

For each triple the producer records the edge-sign pattern in the canonical order $(1, 2), (1, 3), (2, 3)$, the sign product, all three projection coefficients, the minimum target residual, and the normalized volume. The independent checker reconstructs the physical output in reverse column-first order, rather than calling the producer's row builder. Parent code and result hashes, and the frozen TPC-268 engine hash, are locked.

6 Finite atlas

The edge-pattern breakdown is

$$+++ : 5715, \quad +-+ : 1, \quad +-+ : 8, \quad +-- : 3.$$

The two patterns with an odd number of negative edges are exactly the nine anti-alignable cases predicted by Theorem 1. The exceptional cases are confined to the early crossover row $(N, H, Q, z, s) = (256, 38, 27, 5, 1)$; the late growth row $(512, 58, 90, 5, 2)$ contributes all 680 triples with pattern $+++$.

Table 1: Global exact-rational finite census.

quantity	count
rows	18
prime triples	5,727
positive normalized volumes	5,727
zero or negative volumes	0 / 0
anti-alignable triangles	9
sign-frustrated triangles	5,718
minimum target residual $\leq 1/2$	5,313
minimum target residual $\leq 1/4$	4,413
minimum target residual $\leq 1/10$	3,620

Table 2: Selected row summaries; the residual is minimized over the three possible targets.

axis	N	H	Q	s	$ S $	frustrated	min residual
growth	128	24	9	2	3	1	0.560719
growth	192	32	16	2	5	10	0.427252
growth	384	50	40	2	10	120	0.161563
growth	512	58	70	2	15	455	0.025969
growth	512	58	90	2	17	680	0.011508
crossover	256	38	27	1	7	26	0.723005
control	384	52	70	2	15	455	0.018492

The smallest recorded target residual is approximately 0.0115083261 at the triple (167, 173, 179) in the late growth row, with edge pattern + + +. This is simultaneously a strong three-vector linear dependence signal and a sign-frustrated triangle. Geometric near-dependence therefore does not by itself supply a feasible all-pair cancellation sign pattern.

7 Interpretation and claim firewall

The finite experiment closes one uncertainty and exposes another.

1. The exact pairwise cancellation directions from TPC-291 do not compose automatically: cycle parity is already an obstruction at three primes.
2. The obstruction is strongest in the late coherent block, where all tested triangle edges are positive. A signed subtraction can lower a selected pair, but no choice of three coefficient signs lowers all cross terms simultaneously.
3. The nine anti-alignable triangles are localized finite exceptions. A useful next experiment must test whether they form a shell-level signed-graph structure or are merely isolated artifacts of the crossover row.

The exact claims are Theorem 1, Proposition 2, and Gram positivity. The numerical claim is the complete finite atlas on the declared grid. A growing-shell compatibility theorem, source-restricted coefficient image, arithmetic L^2 estimate, fixed-power credit, full Gate B, and a twin-prime conclusion remain open. The Session-named Route-A/Route-B evaluator files are absent from this checkout, so no official evaluator pass is claimed; the local proof, certificate, independent replay, stress test, and Bridge-B checker form the fail-closed validation package.

8 Reproducibility and conclusion

From the project directory run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc292_three_prime_sign_frustration_certificate.py --check
python -B experiments/tpc292_independent_checker.py
python -B experiments/tpc292_frustration_stress.py
```

The canonical result is `results/tpc292_certificate.json`. Normal and optimized runs are required to emit empty standard error and identical standard output.

TPC-292 advances the route map from pairwise cancellation to a precise multi-prime compatibility test. Its main reusable object is the signed Gram graph: cycle parity should be checked before spending effort on coefficient magnitudes or arithmetic endpoint estimates.

References

- [1] Liang Wang. Adaptive shell weighting and the nonnegative coherence wall, 2026. TPC-290 project release in the same research repository.
- [2] Liang Wang. Cross-prime gram coherence in a literal twin-prime dynamics, 2026. TPC-289 project release in the same research repository.
- [3] Liang Wang. Growing-shell gram obstruction in a literal twin-prime dynamics, 2026. TPC-288 project release in the same research repository.
- [4] Liang Wang. Prime-shell cancellation depth in a literal twin-prime dynamics, 2026. TPC-287 project release in the same research repository.
- [5] Liang Wang. Signed schur cancellation in cross-prime gram pairs, 2026. TPC-291 project release in the same research repository.